
LOW-RANK FRICTION FOR MEMORY-EFFICIENT TRANSFORMER PRETRAINING

PREPRINT

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ABSTRACT

iKFAD is a recently proposed optimiser that replaces adaptive learning rates with adaptive friction in the momentum dynamics, yet performs as well as Adam. Its limitation is that the full friction tensor $\xi \in \mathbb{R}^{m \times n}$ carries the same $\mathcal{O}(mn)$ memory overhead per layer as Adam’s second-moment buffer. Here we replace iKFAD’s friction tensor ξ with a rank-1 outer-product factorisation built from row and column momentum statistics, resulting in Rank-1 iKFAD (R-iKFAD). This reduces the friction memory footprint from $\mathcal{O}(mn)$ to $\mathcal{O}(m + n)$ per layer, which approximately halves iKFAD’s total optimiser state. Despite this reduction, R-iKFAD maintains parity in performance with iKFAD: experiments on GPT2-Nano, TinyViT, DistilBERT and GPT2-S confirm that it matches or exceeds iKFAD while nearly halving the memory footprint and remaining comparably robust to hyperparameters. We analyse the continuous-time dynamics in two damping regimes. For linear damping ($\gamma > 0$) we prove exponential convergence under strong convexity. For $\gamma = 0$, the preferred option in our experiments, the friction is generated entirely from past momentum and switches off as the momentum vanishes, so geometric convergence cannot be shown. We nonetheless prove convergence to the minimiser, together with matching upper and lower bounds on the energy: of order t^{-1} when the regularisation scale ϵ_{stab} is zero, and of order $t^{-1/2}$ when it is positive. To our knowledge this is the first convergence rate for a rank-1 factored optimiser in continuous time, and the first such result that does not require positive damping.

1 Introduction

Adam [9] is the workhorse optimiser in modern deep learning, combining gradient momentum with coordinate-wise second-moment scaling to accelerate convergence on complex loss landscapes. For a model with N scalar parameters, Adam evolves two auxiliary states (the first and second moment estimates) and therefore requires $2N$ auxiliary scalars on top of the N parameters themselves, for a total memory cost of around $3N$. At scale, this $2N$ overhead presents a significant hardware bottleneck during large-model training. To mitigate this, Adafactor [16] introduces a rank-1 factorisation of the second-moment matrix. For a matrix weight $W \in \mathbb{R}^{m \times n}$, Adafactor replaces Adam’s full second-moment matrix $V \in \mathbb{R}^{m \times n}$ with a factorised approximation $\hat{V}_{ij} = R_i C_j / (\mathbf{1}_m^\top R)$, where $R \in \mathbb{R}^m$ and $C \in \mathbb{R}^n$ track row-wise and column-wise squared gradients, respectively. By eliminating the first-moment buffer, Adafactor reduces per-layer auxiliary state complexity from $\mathcal{O}(mn)$ to $\mathcal{O}(m + n)$ (achieving memory savings of over 99% for large weight matrices) while maintaining competitive transformer pretraining performance.

An insightful paradigm for analysing optimiser dynamics is based on continuous-time ordinary differential equations (ODEs) [3, 5], where updates emerge as discrete time-step approximations. For parameter state x , momentum p , and second-moment variable ζ , continuous-time Adam is given in Eqs. (1)-(3). While Adam incorporates parameter-wise adaptation into the *position* equation ($\dot{x}$) via $1/\sqrt{\zeta}$, Karoni et al. [7] demonstrated that coordinate-wise adaptation can alternatively be integrated into the *momentum* equation ($\dot{p}$). This is achieved by replacing the constant damping factor γ

with a dynamic, coordinate-wise friction state ξ , resulting in the Individual Kinetic Friction-Adaptive Descent (iKFAD) dynamics (4)-(6)

$$\dot{x} = \frac{p}{\sqrt{\zeta} + \epsilon}, \quad (1)$$

$$\dot{p} = -\nabla f(x) - \gamma p, \quad (2)$$

$$\dot{\zeta} = [\nabla f(x)]^2 - \alpha \zeta. \quad (3)$$

$$\dot{x} = p, \quad (4)$$

$$\dot{p} = -\nabla f(x) - (\gamma + \xi) \odot p, \quad (5)$$

$$\dot{\xi} = \frac{[p]^2}{\mu} - \alpha \xi. \quad (6)$$

These dynamics require storing a full adaptive friction tensor ξ alongside momentum p . They incur an auxiliary footprint of $2N$ scalars, identical to Adam’s auxiliary storage. Originally introduced for scalar/global friction [6], iKFAD extends friction-based adaptation to individual parameters, matching Adam’s performance on standard vision and NLP benchmarks. This leaves an evident gap. Adafactor obtains its saving by factorising the object Adam stores in the position equation, yet no counterpart exists for the object iKFAD stores in the momentum equation. Figure 1 arranges the four methods on the two axes that organise this paper: which equation the adaptation enters, and whether the adapted object is held in full or in rank-1 factored form. The lower-right cell is the one we fill. Rank-1 iKFAD (R-iKFAD) applies Adafactor’s factorisation to the friction tensor, reducing iKFAD’s auxiliary state from $2N$ to $N + \sum_{\ell}(m_{\ell} + n_{\ell})$ scalars while retaining full momentum.

2 Methodology: Rank-1 iKFAD

Adafactor compresses Adam’s second moment; Rank-1 iKFAD (R-iKFAD) applies the same compression to iKFAD’s friction. Writing both as continuous-time systems for a matrix weight $X \in \mathbb{R}^{m \times n}$ makes the correspondence apparent, in the same sense in which the systems above approximate Adam and iKFAD [3, 5]. Most Adafactor implementations omit the first moment, but we enable it throughout for parity, so we retain momentum P and refer to the result as **Adafactor-m** (left). R-iKFAD (right) carries the same momentum and factorises the friction instead:

$$\dot{X} = \frac{P}{\sqrt{\hat{\zeta}} + \epsilon}, \quad (7)$$

$$\dot{P} = -\nabla f(X) - \gamma P, \quad (8)$$

$$\dot{R} = [\nabla f(X)]^2 \mathbf{1}_n - \alpha R, \quad (9)$$

$$\dot{C} = ([\nabla f(X)]^2)^{\top} \mathbf{1}_m - \alpha C. \quad (10)$$

$$\dot{X} = P, \quad (11)$$

$$\dot{P} = -\nabla f(X) - \tilde{\xi} \odot P - \gamma P, \quad (12)$$

$$\dot{R} = \frac{1}{\mu} [P]^2 \mathbf{1}_n - \alpha R, \quad (13)$$

$$\dot{C} = \frac{1}{\mu} ([P]^2)^{\top} \mathbf{1}_m - \alpha C. \quad (14)$$

Here $\odot$ is the element-wise product, $[P]_{ij}^2 = P_{ij}^2$, and the divisions in the position equations are element-wise; both systems carry the same linear damping γ . Each builds a rank-1 object from its factors,

$$\hat{\zeta} = \frac{RC^{\top}}{\mathbf{1}_m^{\top} R}, \quad \tilde{\xi}_{ij} = \frac{R_i C_j}{\mathbf{1}_m^{\top} R + \epsilon_{\text{stab}}}, \quad (15)$$

with $\epsilon, \epsilon_{\text{stab}} > 0$ small regularisers, the latter ensuring numerical stability near $R = 0$. Adafactor-m is thus Adam’s system with the full second moment ζ replaced by the rank-1 estimate $\hat{\zeta}$, and R-iKFAD is iKFAD’s system with the full friction tensor ξ replaced by the rank-1 estimate $\tilde{\xi}$. The two differ in the same way Adam differs from iKFAD: Adafactor-m’s factors aggregate squared *gradients* and act in the position equation, whereas R-iKFAD’s aggregate squared *momentum* and act as damping in the momentum equation. In both cases the state per layer falls from $\mathcal{O}(mn)$ to $\mathcal{O}(m + n)$ while full momentum is retained. We emphasise that R-iKFAD is designed to maintain parity with iKFAD (which matches Adam [7] in performance), rather than outperforming it.

Splitting Discretisation. To construct a stable discrete-time algorithm, we rely on principles of geometric numerical integration of Hamiltonian systems [11], which have proved a robust foundation for discretisation of dynamics-based optimisation schemes. Following [7], we decompose the vector field of (11)–(14) into four sub-operators whose individual flows can be integrated analytically:

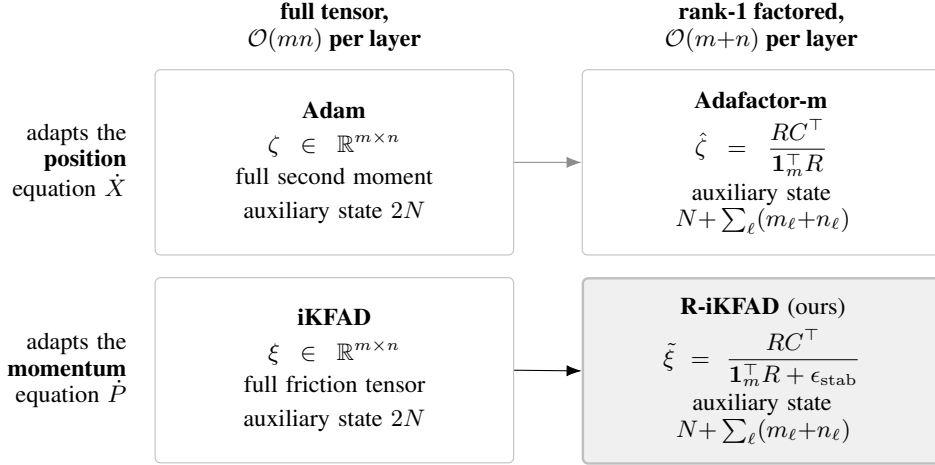


Figure 1: The same rank-1 compression, applied in two different equations. Columns distinguish whether the adapted object is stored in full or as a rank-1 outer product of the factors (R, C) ; rows distinguish which equation it acts in. Adafactor-m factors Adam’s second moment ζ , which scales the *position* update, whereas R-iKFAD factors iKFAD’s friction ξ , which damps the *momentum* update. Both retain full momentum and reduce per-layer auxiliary state from $\mathcal{O}(mn)$ to $\mathcal{O}(m+n)$, so the total auxiliary footprint falls from $2N$ to $N + \sum_{\ell} (m_{\ell} + n_{\ell}) \approx N$. Non-2D parameters (biases, LayerNorm scales) are not rank-1 factorisable and fall back to element-wise iKFAD.

$$\begin{pmatrix} \dot{X} \\ \dot{P} \\ \dot{R} \\ \dot{C} \end{pmatrix} = \underbrace{\begin{pmatrix} P \\ 0 \\ 0 \\ 0 \end{pmatrix}}_A + \underbrace{\begin{pmatrix} 0 \\ -\nabla f(X) \\ 0 \\ 0 \end{pmatrix}}_B + \underbrace{\begin{pmatrix} 0 \\ -\frac{RC^T}{\mathbf{1}_m^T R + \epsilon_{\text{stab}}} \odot P \\ \frac{1}{\mu} [P]^2 \mathbf{1}_n - \alpha R \\ \frac{1}{\mu} ([P]^2)^T \mathbf{1}_m - \alpha C \end{pmatrix}}_C + \underbrace{\begin{pmatrix} 0 \\ -\gamma P \\ 0 \\ 0 \end{pmatrix}}_D. \quad (16)$$

Sub-flows A, B, and D integrate analytically: A gives position drift; B applies the gradient kick; D applies linear friction (becoming the identity when $\gamma = 0$, i.e., R-iKFAD0). Sub-flow C couples momentum P and friction factors (R, C) and is integrated semi-analytically via an internal symmetric freeze-then-update scheme. The composition applied in practice is a first-order Lie–Trotter splitting:

$$\Phi_h = \Phi_h^D \circ \Phi_h^C \circ \Phi_h^A \circ \Phi_h^B,$$

where sub-steps are executed in the sequence $B \rightarrow A \rightarrow C \rightarrow D$ (BACD) [7]. Position updates use the gradient-kicked momentum $P^{(B)}$ rather than P_n . Table 3 in Appendix A gives the explicit analytical update rules for each sub-step across the complete BACD sequence over time step h , together with the treatment of the coupling inside sub-flow C.

Memory Complexity. For a network parameter matrix at layer ℓ given by $W_{\ell} \in \mathbb{R}^{m_{\ell} \times n_{\ell}}$, let $N = \sum_{\ell} m_{\ell} n_{\ell}$ denote the total number of scalar model parameters. Figure 1 breaks down the auxiliary optimiser memory footprint required on top of the model weights. For non-2D parameters (biases, LayerNorm scales) that are not rank-1 factorisable, R-iKFAD falls back to element-wise iKFAD, contributing a negligible additional scalar per parameter. R-iKFAD requires approximately N auxiliary scalars in total, effectively halving the optimiser auxiliary memory footprint relative to both iKFAD and Adam. Adafactor-m occupies this same $\approx N$ regime and is therefore our primary baseline; the difference is which buffer is compressed, the second moment for Adafactor-m versus the friction tensor for R-iKFAD. Retaining momentum is deliberate: Adafactor is known to suffer training instabilities in certain transformer regimes when momentum is discarded entirely [16].

3 Theoretical Results

We analyse the continuous-time system (11)–(14) in both damping regimes. This section states the results and reports the numerical verification; the proofs are in Appendix G.

For $\gamma > 0$, a Lyapunov argument gives exponential convergence under strong convexity.

Theorem 1 (Exponential convergence for $\gamma > 0$). *Suppose that f is m_f -strongly convex and has an L_f -Lipschitz gradient. Let $\gamma, \alpha, \mu, \epsilon_{\text{stab}} > 0$, and let $R(0)$ and $C(0)$ be componentwise nonnegative. Then there are constants $M, \rho > 0$ such that*

$$f(X(t)) - f(X^*) + \|X(t) - X^*\|_F^2 + \|P(t)\|_F^2 + \|R(t)\|^2 + \|C(t)\|^2 \leq M e^{-\rho t} \quad (17)$$

for every $t \geq 0$.

Our experiments use $\gamma = 0$. In that case the adaptive friction remains non-negative but can become small with the momentum. We prove convergence at $\gamma = 0$ under the assumptions of the next theorem.

Theorem 2 (Convergence when $\gamma = 0$). *Suppose that f is strongly convex and has a locally Lipschitz gradient. Let $\alpha, \mu, \epsilon_{\text{stab}} > 0$, and suppose that $R(0)$ and $C(0)$ are componentwise nonnegative. Then every solution of (11)–(14) with $\gamma = 0$ converges to*

$$(X, P, R, C) = (X_*, 0, 0, 0),$$

where X_* is the unique minimiser of f .

The decay at $\gamma = 0$ is algebraic rather than exponential, and which exponent applies is decided by ϵ_{stab} . Both this theorem and the numerical evidence that follows are stated in terms of the energy

$$\mathcal{E}(t) = f(X(t)) - f(X^*) + \frac{1}{2} \|P(t)\|_F^2, \quad (18)$$

the sum of the objective gap and the kinetic term, which is nonnegative and vanishes exactly at $(X^*, 0)$.

Theorem 3 (Algebraic rates). *Suppose that f is strongly convex, its gradient is locally Lipschitz, and it is C^2 in a neighbourhood of X^* . Let $\alpha, \mu > 0$, and let the initial factors be componentwise nonnegative.*

If $\epsilon_{\text{stab}} > 0$ and $\mathcal{E}(0) > 0$, there are $a, A > 0$ such that

$$\frac{a}{\sqrt{1+t}} \leq \mathcal{E}(t) \leq \frac{A}{\sqrt{1+t}}, \quad t \geq 0.$$

If $\epsilon_{\text{stab}} = 0$, assume in addition that $R(0) = C(0) = 0$, and use the convention $\tilde{\xi} = 0$ at $\mathbf{1}_m^\top R = 0$. If $\mathcal{E}(0) > 0$, there are $b, B > 0$ such that

$$\frac{b}{1+t} \leq \mathcal{E}(t) \leq \frac{B}{1+t}, \quad t \geq 0.$$

The two regimes are separated by a crossover when $\mathcal{E}$ is of order $\alpha\mu\epsilon_{\text{stab}}$. Section 3.1 gives a formal averaging calculation that identifies the exponents, Appendix G.3.1 proves them, and Section 3.2 verifies both numerically. Because the bounds are two-sided, $\epsilon_{\text{stab}} > 0$ does not merely permit the slower $t^{-1/2}$ decay but forces it, which is a reason to keep ϵ_{stab} small.

3.1 Formal rate calculation

We use a formal averaging calculation to identify the decay exponents. The exponents it produces are established rigorously in Appendix G.3.1. Suppose that P oscillates faster than the factors change. When $\epsilon_{\text{stab}} = 0$, begin after $\sum_k R_k > 0$. Replacing the oscillatory terms by their averages gives

$$\dot{\mathcal{E}} \simeq -\frac{\mathcal{E}^2}{\alpha\mu}.$$

The reduced equation has the solution

$$\mathcal{E}(t) \simeq \frac{\alpha\mu}{t}.$$

When $\epsilon_{\text{stab}} > 0$ and the factors are small relative to ϵ_{stab} , the same calculation gives

$$\dot{\mathcal{E}} \simeq -\frac{\mathcal{E}^3}{\alpha^2\mu^2\epsilon_{\text{stab}}}, \quad \mathcal{E}(t) \simeq \alpha\mu\sqrt{\frac{\epsilon_{\text{stab}}}{2t}}.$$

The two reduced regimes meet when $\mathcal{E}$ is of order $\alpha\mu\epsilon_{\text{stab}}$. This calculation uses a time-scale separation and an averaging closure. Neither assumption is needed for Theorem 3, which reaches the same exponents without them.

Table 1: Local slopes of $\log \mathcal{E}$ against $\log t$ on successive decades, from the BACD splitting at $h = 0.02$ with $\alpha = \mu = 1$, integrated to $T = 10^7$. Both exponents are independent of the matrix dimensions.

	$[10^3, 10^4]$	$[10^4, 10^5]$	$[10^5, 10^6]$	$[10^6, 10^7]$
$\epsilon_{\text{stab}} = 0$ (proved exponent -1)				
$m = n = 2$	-0.955	-0.980	-0.986	-0.989
$m = 4, n = 3$	-0.991	-0.996	-0.996	-0.998
$m = 6, n = 5$	-0.993	-0.997	-0.998	-0.999
$\epsilon_{\text{stab}} = 0.1$ (proved exponent $-1/2$)				
$m = n = 2$	-0.499	-0.497	-0.496	-0.497
$m = 4, n = 3$	-0.558	-0.518	-0.505	-0.501
$m = 6, n = 5$	-0.593	-0.530	-0.509	-0.503

3.2 Numerical evidence

We integrate the $\gamma = 0$ dynamics on a strongly convex quadratic, $f(X) = \frac{1}{2}X^\top AX$ with $A = BB^\top/d + 0.8I$ and $d = mn$, where B has independent standard normal entries, so the smallest eigenvalue of A is at least 0.8, and we track the energy $\mathcal{E}(t) = \frac{1}{2}X^\top AX + \frac{1}{2}\|P\|_F^2$. Figure 2 shows the decay at both stabiliser settings and Table 1 the local slopes of $\log \mathcal{E}$ against $\log t$, obtained by ordinary least squares on successive decades out to $T = 10^7$ at three matrix sizes. Both proved exponents are recovered and neither depends on the matrix dimensions. The approach to $-1/2$ is slower for larger mn , and the crossover explains why: the second regime begins only once $\mathcal{E} \lesssim \alpha\mu\epsilon_{\text{stab}}$, and the constant in the first-regime law grows with mn , so the crossing occurs later for larger systems. The prefactors scale with α, μ and ϵ_{stab} as the calculation of Section 3.1 predicts; Appendix E reports the measured ratios and Appendix F the integrators, tolerances, step sizes, seeds and initialisation.

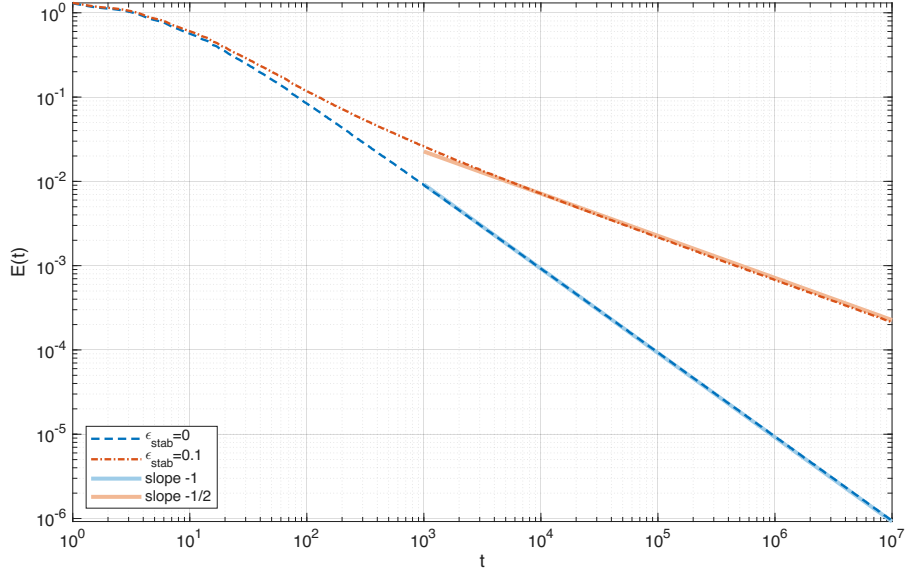


Figure 2: Energy decay from BACD splitting with $m = 4, n = 3, \gamma = 0, \alpha = \mu = 1, h = 0.02$, and final time 10^7 . Both runs have $R(0) = C(0) = 0$. The blue dashed curve has $\epsilon_{\text{stab}} = 0$, and the orange dash-dotted curve has $\epsilon_{\text{stab}} = 0.1$. The solid lines have slopes -1 and $-1/2$, respectively.

3.3 Related Work: Algebraic Rates and Factored Optimisers

Continuous-time analyses with damping prescribed as a function of time can give algebraic rates. Examples include Su et al. [18] and Attouch et al. [2]. Those results do not directly apply here because our damping is determined by evolving factors and can vanish with them. On the factored side, Adafactor [16] introduced rank-1 factorisation of optimiser state. Hong and Lin [4] analyse a discrete Adafactor method for nonconvex objectives. Nguyen et al. [13] introduce H-Fac, which factorises both the momentum and scaling estimators, and prove asymptotic convergence for

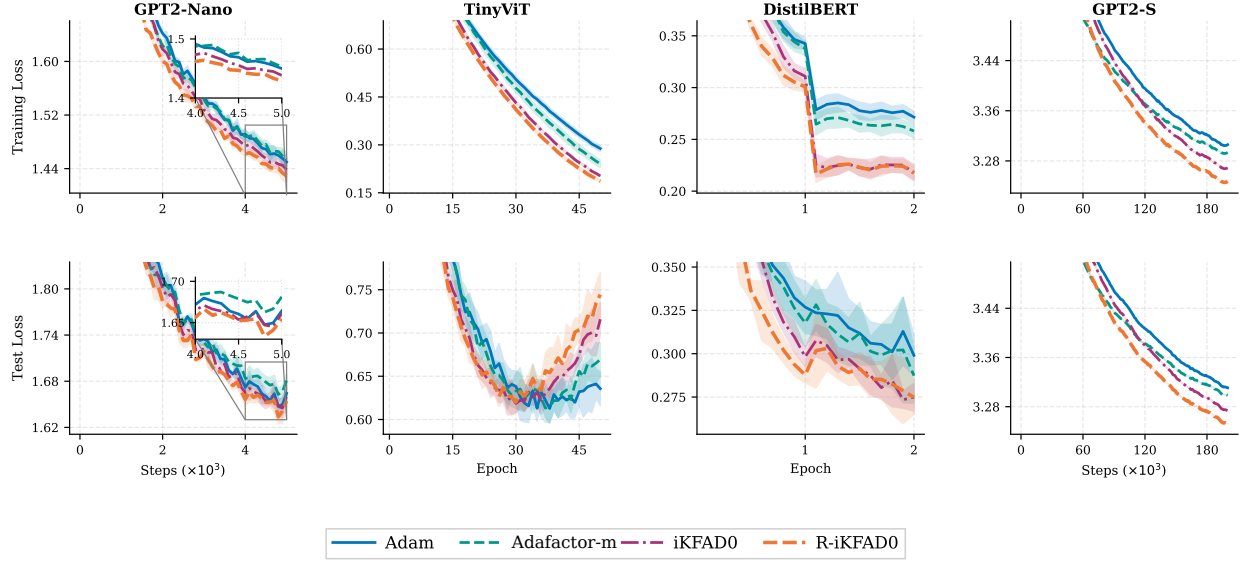


Figure 3: Loss curves for all four benchmarks, averaged over 10 random seeds. R-iKFAD0 matches or improves upon iKFAD0 on all tasks, while using approximately half the optimiser memory. Both methods are competitive with Adam and Adafactor-m, which require comparable or greater memory. Shaded regions denote one standard deviation.

its continuous-time dynamics with positive damping. These results do not establish the $\gamma = 0$ rates proved above. R-iKFAD builds on scalar friction [6] and full-tensor iKFAD [7]. For $\gamma = 0$ we prove continuous-time convergence together with matching upper and lower algebraic rates (Theorem 3). The numerical results in Section 3.2 agree with both proved exponents.

4 Experiments

We set $\gamma = 0$ for both iKFAD and R-iKFAD throughout (denoted **iKFAD0** and **R-iKFAD0** respectively), following [7]. In the ablation in Section 4.2, adding $\gamma > 0$ gave no consistent improvement under the tested procedure. We compare both against Adam and Adafactor-m across four benchmarks spanning image classification and language modelling at scales from 0.85M to 125M parameters:

- **GPT2-Nano (0.85M)** [8] on Shakespeare character-level language modelling.
- **TinyViT (5M)** [19] on CIFAR-10 [10] image classification.
- **DistilBERT (80M)** [15] on SST-2 [17] sentiment classification (pretraining from scratch).
- **GPT2-S (125M)** [14] on OpenWebText pretraining (nanoGPT architecture) [8].

To isolate the optimisation dynamics, we use a constant learning rate, disable weight decay, and omit additional regularisers throughout. We use Adafactor-m rather than the conventional momentum-free configuration because its auxiliary-memory requirement is similar to that of R-iKFAD. All hyperparameters are tuned via Optuna [1] (TPE sampler) with equal sweep budgets of approximately 80 trials per optimiser per model; GPT2-Nano and TinyViT received substantially more trials. The stabiliser is fixed at $\epsilon_{\text{stab}} = 10^{-16}$ in every run and is never tuned. See Appendix B for full configurations. As with iKFAD, the optimal scale of μ varies by several orders of magnitude between tasks due to differing momentum scales. We therefore recommend a wide-range initial sweep over μ , and accept this as a limitation. Note that Adafactor-m requires tuning only two hyperparameters, whereas Adam, iKFAD0, and R-iKFAD0 all require three. The fixed trial budget hence favoured Adafactor-m’s results disproportionately. All benchmarks report mean $\pm$ standard deviation over 10 seeds, except GPT2-S, which uses a single seed due to computational constraints. The optimiser auxiliary state is the memory an optimiser adds on top of the model parameters; Table 5 in Appendix C reports the absolute values on all four benchmarks in MB (float32).

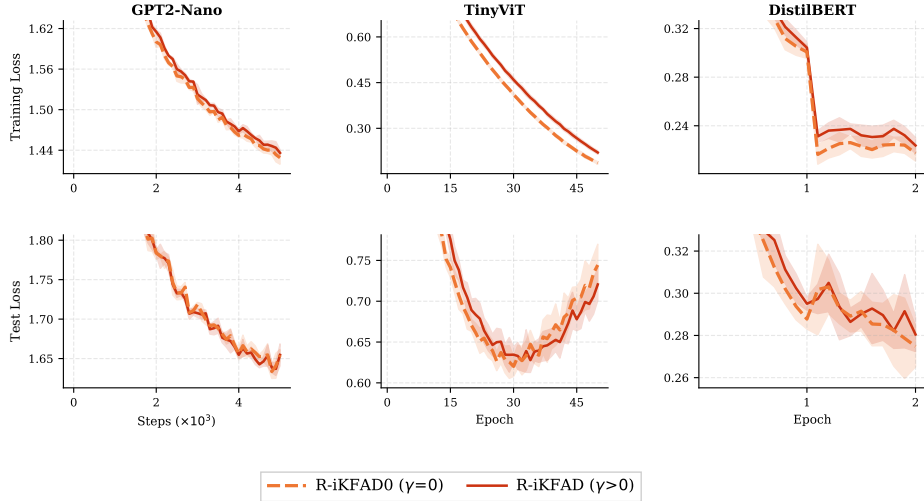


Figure 4: Training and test loss for R-iKFAD ($\gamma > 0$) vs. R-iKFAD0 ($\gamma = 0$) on GPT2-Nano, TinyViT, and DistilBERT (SST-2), averaged over 10 seeds (shaded = std). The two variants are indistinguishable in performance, supporting R-iKFAD0 as the recommended default.

4.1 Main Results

Figure 3 shows training and test loss across all benchmarks, and Figure 7 in Appendix D the corresponding accuracies. R-iKFAD0 gave results similar to iKFAD0 while using about half the optimiser memory. It also gave results comparable to Adam in these experiments, consistent with the results for iKFAD in [7]. The aim is to reduce iKFAD’s memory requirement while retaining similar empirical performance. Table 2 summarises the test loss at the validation-selected checkpoint for all methods. The margins deserve care. On the three benchmarks with seed replication, iKFAD0 and R-iKFAD0 differ by at most half a pooled standard deviation, and on TinyViT they agree to three decimal places, so at this resolution the factorisation costs nothing measurable. Where the friction methods do separate from Adam, the direction depends on the domain: on DistilBERT they lead by about four pooled standard deviations, and on GPT2-S by 0.06 nats at a single seed, whereas on the one vision task Adam leads by less than one. Given how small these differences are, we do not claim that R-iKFAD0 outperforms Adam; the claim is the narrower one that the friction tensor can be rank-1 factorised without compromising performance.

Table 2: Test loss at the checkpoint selected by lowest *validation* loss; the test set was never used for model or hyperparameter selection (mean $\pm$ std over 10 seeds; GPT2-S: single seed due to computational limitations). Lower is better. R-iKFAD0 matches or improves upon iKFAD0 across all benchmarks while using $\approx 2\times$ less optimiser memory.

Dataset	Adam	Adafactor-m	iKFAD0	R-iKFAD0 (ours)
GPT2-Nano	1.643 ± 0.008	1.661 ± 0.007	1.640 ± 0.008	1.633 ± 0.011
TinyViT	0.593 ± 0.015	0.596 ± 0.010	0.606 ± 0.011	0.606 ± 0.012
DistilBERT	0.292 ± 0.005	0.280 ± 0.007	0.269 ± 0.005	0.268 ± 0.003
GPT2-S	3.301	3.287	3.263	3.242

4.2 Ablation: $\gamma = 0$ versus $\gamma > 0$

Figure 4 compares R-iKFAD ($\gamma > 0$, with linear damping) against R-iKFAD0 ($\gamma = 0$, without linear damping) on GPT2-Nano, TinyViT, and DistilBERT (SST-2). Adding the γ term gave no consistent improvement over $\gamma = 0$ on these benchmarks under the tested procedure. This is consistent with the finding in [7] that the adaptive friction mechanism alone provides sufficient damping. We therefore recommend **R-iKFAD0** ($\gamma = 0$) as the practical default since it has one fewer hyperparameter. For the $\gamma > 0$ variant, we used the same h , α , and μ settings as R-iKFAD0 given in Appendix B and additionally swept γ log-uniformly over $[10^{-6}, 10]$.

4.3 Robustness to Hyperparameters

Figure 5 shows γ - h grid sweeps for R-iKFAD and iKFAD. Both methods have broad regions of the tested grid with similar performance. In the R-iKFAD equations, an increase in momentum increases the factors R and C , which increases $\tilde{\xi}$ and hence the damping. In these sweeps, the rank-1 approximation did not substantially reduce the range of (γ, h) values giving low loss. The breadth of the plateau follows from that feedback being negative: a step size large enough to inflate the momentum also inflates R and C , and so $\tilde{\xi}$, so the damping rises to meet it. The same mechanism explains why γ adds little in Section 4.2: a constant damping term acts along a direction the adaptive term already covers, but without responding to the state.

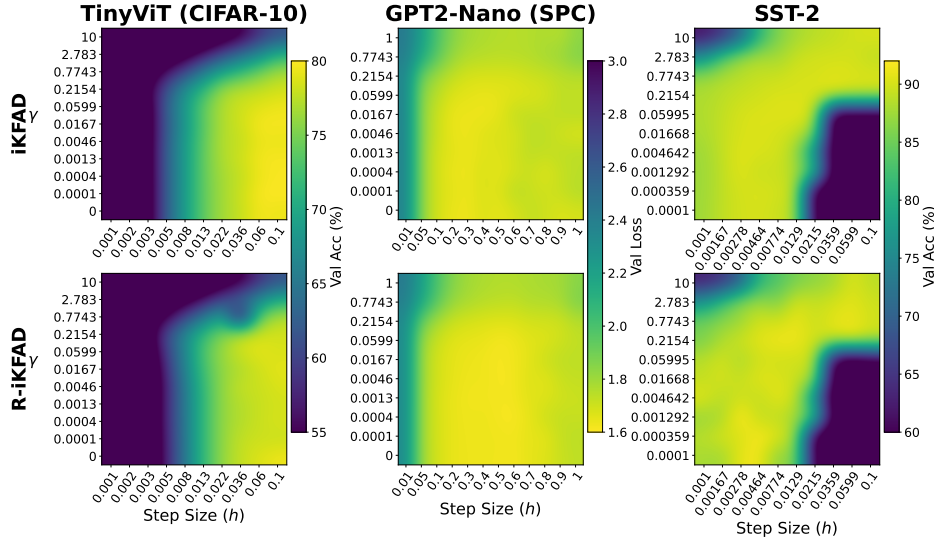


Figure 5: Two-dimensional γ - h sensitivity for R-iKFAD (top) and iKFAD (bottom). Each cell shows best performance at a fixed (h, γ) grid point; brighter = better. R-iKFAD retains comparably broad high-performance plateaus.

4.4 How Close is $\tilde{\xi}$ to the Best Rank-1 Friction?

Two separate things could limit the factorisation: ξ might not be close to rank-1, or the cheap construction might be a poor rank-1 approximation even when a good one exists. The Eckart-Young theorem separates them, since the smallest relative error achievable by *any* rank-1 matrix is $\sqrt{1 - \sigma_1^2 / \sum_i \sigma_i^2}$. Figure 6 reports both. To compare ξ and $\tilde{\xi}$ on identical iterates we run iKFAD0 while carrying R-iKFAD’s factors (R_ℓ, C_ℓ) as read-only shadow state, leaving the optimiser’s own updates untouched. Across all four benchmarks $\tilde{\xi}$ lands between 0.07 and 0.85 percentage points above the floor; on GPT2-S the median excess over the 74 two-dimensional layers is 0.06 pp, with 61 of them within 1 pp of optimal. The largest excesses fall on the layers where ξ is *most* rank-1, because the floor tends to zero there. The stable rank $\|\xi\|_F^2 / \|\xi\|_2^2$ runs from 1.18 on TinyViT to 2.01 on GPT2-Nano, and within GPT2-S is organised by module rather than randomly: early attention value projections approach 1, while mid-network query and key projections sit furthest from it. The error R-iKFAD pays is therefore almost entirely attributable to ξ not being exactly rank-1 rather than to the construction being crude. This is the empirical counterpart of the parity reported in Section 4.1.

5 Conclusion

We introduced **R-iKFAD**, which replaces iKFAD’s full friction tensor with a rank-1 outer product of momentum row and column sums, reducing per-layer friction state from $\mathcal{O}(mn)$ to $\mathcal{O}(m+n)$ while retaining a full momentum buffer. Across GPT2-Nano, TinyViT, DistilBERT and GPT2-S, R-iKFAD matches iKFAD in convergence speed, final performance and hyperparameter robustness at roughly half the auxiliary optimiser memory. The accompanying analysis, unlike the usual treatment of momentum-based dynamics, does not require positive linear damping. Under strong convexity we prove exponential convergence when $\gamma > 0$ and convergence to the minimiser when $\gamma = 0$. In the undamped case we prove matching upper and lower bounds on the energy, of order t^{-1} when $\epsilon_{\text{stab}} = 0$ and of order $t^{-1/2}$ when $\epsilon_{\text{stab}} > 0$, so a constant introduced only to avoid a division by zero is what fixes the asymptotic rate.

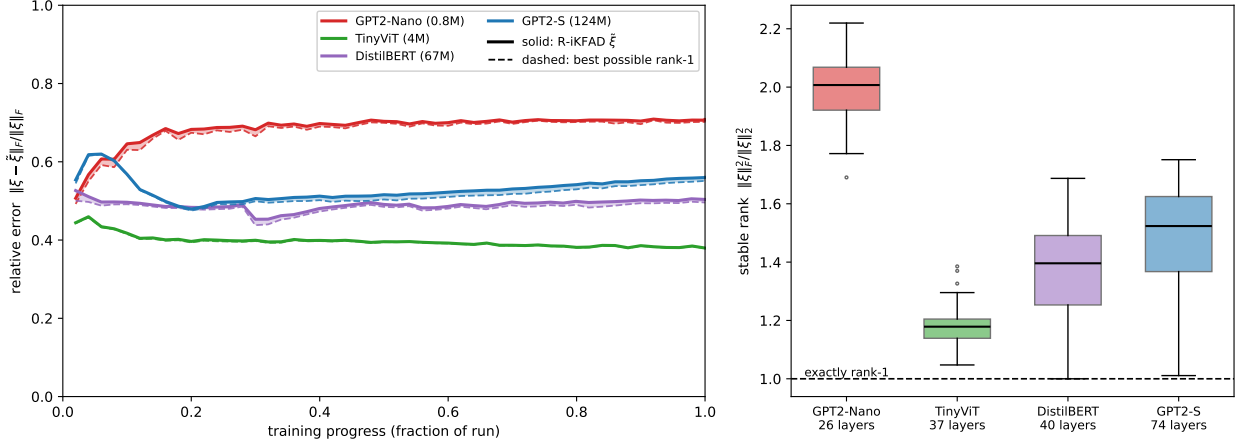


Figure 6: R-iKFAD’s rank-1 friction against the best rank-1 approximation of the same tensor. **Left:** layer-averaged relative error $\|\xi - \tilde{\xi}\|_F / \|\xi\|_F$ (solid) against the Eckart–Young floor $\sqrt{1 - \sigma_1^2 / \sum_i \sigma_i^2}$ (dashed); the shaded gap is the excess attributable to the construction, and is never wider than 0.85 pp. The horizontal axis is the fraction of each run completed, as the four benchmarks use different step budgets. **Right:** distribution over layers of the stable rank of ξ at the final step, where 1 would be exactly rank-1. Measured along iKFAD0 trajectories carrying R-iKFAD’s factors as read-only shadow state.

Numerical integration of the dynamics agrees with both exponents. The analysis is continuous-time, deterministic and assumes strong convexity. Discrete-time and stochastic-gradient results remain open.

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A Update Rules for Splitting Discretisation

This appendix gives the explicit sub-step updates for the BACD splitting introduced in Section 2, and the treatment of the one sub-flow that does not admit a closed-form joint solution.

Table 3: Analytical sub-step update rules for the BACD splitting sequence over time step h .

Sub-step	Update Equation
B	$P^{(B)} = P_n - h \nabla f(X_n)$
A	$X_{n+1} = X_n + h P^{(B)}$
C	$\tilde{\xi}_n = \frac{R_n C_n^\top}{\mathbf{1}_m^\top R_n + \epsilon_{\text{stab}}}$ $P^{(1)} = P^{(B)} \odot \exp\left(-\frac{h}{2} \tilde{\xi}_n\right)$ $R_{n+1} = e^{-\alpha h} R_n + \frac{1 - e^{-\alpha h}}{\mu \alpha} [P^{(1)}]^2 \mathbf{1}_n$ $C_{n+1} = e^{-\alpha h} C_n + \frac{1 - e^{-\alpha h}}{\mu \alpha} ([P^{(1)}]^2)^\top \mathbf{1}_m$ $\tilde{\xi}_{n+1} = \frac{R_{n+1} C_{n+1}^\top}{\mathbf{1}_m^\top R_{n+1} + \epsilon_{\text{stab}}}$ $P^{(2)} = P^{(1)} \odot \exp\left(-\frac{h}{2} \tilde{\xi}_{n+1}\right)$
D	$P_{n+1} = P^{(2)} \odot e^{-\gamma h} \quad (\text{identity when } \gamma = 0)$

Within sub-flow C, no closed-form joint solution exists for coupled P and (R, C) . We apply a symmetric half-damp sequence: evaluating $\exp(-\frac{h}{2} \tilde{\xi})$ twice around an exact exponential ODE factor update improves integration accuracy to $\mathcal{O}(h^2)$ within sub-flow C alone. Using the post-first-half-damp momentum $P^{(1)}$ for factor updates reduces forcing magnitude and prevents over-accumulation.

B Optimal Hyperparameters

Across models and tasks, the best-performing scales for μ (iKFAD0, R-iKFAD0) varied by several orders of magnitude. This variation is likely related to the scale of the momentum variables during training, since μ controls the strength of the adaptive friction terms. Normalising the momentum variables across iterations could potentially reduce this sensitivity, but we leave this for future work.

Hyperparameters were selected by minimum validation loss during the Optuna sweeps; all reported results are computed on held-out test data.

Experimental configurations:

- **TinyViT – CIFAR-10:** batch size 128, 25 epochs, 100 trials. $h \in [10^{-6}, 10^{-1}]$, $\alpha \in [10^{-3}, 1]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and R-iKFAD0.
- **DistilBERT – SST-2:** batch size 16, 2 epochs, 80 trials. $h \in [10^{-6}, 10^{-1}]$, $\alpha \in [10^{-3}, 1]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and R-iKFAD0.
- **GPT2-Nano – Shakespeare:** batch size 16, 5000 steps for the hyperparameter sweeps and final reported runs, 500 trials. $h \in [10^{-6}, 5 \times 10^{-1}]$, $\alpha \in [10^{-5}, 10]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and R-iKFAD0.
- **GPT2-S – OpenWebText:** batch size 16, 75001 steps for the hyperparameter sweeps and 200001 steps for the final reported runs, 80 trials. $h \in [10^{-6}, 10^{-1}]$, $\alpha \in [10^{-3}, 1]$, $\mu \in [10^{-8}, 10]$ for iKFAD0 and $\mu \in [10^{-9}, 10^{-2}]$ for R-iKFAD0.

Table 4: Optimised hyperparameters for all experiments. All iKFAD0 and R-iKFAD0 entries have $\gamma = 0$. Dashes indicate parameters not used by the method.

	h	α	μ	β_1	β_2
TinyViT (CIFAR-10)					
iKFAD0	0.0725	0.0899	1.2643×10^{-5}	–	–
R-iKFAD0	0.0860	0.5327	9.4765×10^{-7}	–	–
Adam	5.5311×10^{-4}	–	–	0.8608	0.8852
Adafactor-m	5.6050×10^{-4}	–	–	0.8907	–
DistilBERT (SST-2)					
iKFAD0	0.0165	0.0574	9.8948×10^{-6}	–	–
R-iKFAD0	0.0423	0.0060	8.1874×10^{-7}	–	–
Adam	3.82×10^{-5}	–	–	0.9187	0.9825
Adafactor-m	2.5638×10^{-5}	–	–	0.9622	–
GPT2-Nano (Shakespeare)					
iKFAD0	0.4941	2.1955	4.5521×10^{-6}	–	–
R-iKFAD0	0.4842	2.8475	1.9407×10^{-6}	–	–
Adam	1.6803×10^{-3}	–	–	0.8876	0.9265
Adafactor-m	2.9923×10^{-3}	–	–	0.8893	–
GPT2-S (OpenWebText)					
iKFAD0	0.0938	0.1933	4.2726×10^{-7}	–	–
R-iKFAD0	0.0918	0.0579	1.3637×10^{-6}	–	–
Adam	1.0478×10^{-4}	–	–	0.9000	0.9971
Adafactor-m	7.2777×10^{-4}	–	–	0.8865	–

For all experiments, Adam was swept over $h \in [10^{-6}, 10^{-2}]$, $\beta_1 \in [0.85, 0.999]$, $\beta_2 \in [0.85, 0.999]$. Adafactor-m was swept over $h \in [10^{-5}, 10^{-2}]$ and $\beta_1 \in [0.85, 0.999]$, with no relative step or warmup. Adafactor-m has no tunable β_2 : as originally proposed [16] and in standard practice, the second-moment decay follows the prescribed schedule $\beta_{2,t} = 1 - t^{-0.8}$ (decay_rate = -0.8), which is why its β_2 entries are left blank. We enable $\beta_1 > 0$ rather than the conventional momentum-free setting so that Adafactor-m occupies the same auxiliary-memory regime as R-iKFAD

(Table 5). All sweeps used Optuna’s [1] TPE sampler. Table 4 reports the best hyperparameters found. iKFAD0 and R-iKFAD0 are the $\gamma = 0$ variants; all such entries satisfy $\gamma = 0$ by definition. For iKFAD0 and R-iKFAD0, h denotes the step size and μ the adaptive damping coefficient. For Adam and Adafactor-m, h is the learning rate.

C Memory Analysis

Table 5: Optimiser auxiliary state (MiB, float32), computed over all model parameters. R-iKFAD achieves approximately $2\times$ savings over both iKFAD and Adam because the rank-1 friction factors (R_ℓ, C_ℓ) are negligible relative to the full momentum buffer. The Adafactor-m baseline used throughout enables its optional first moment ($\beta_1 > 0$; Section 4) and therefore carries an auxiliary footprint identical to R-iKFAD’s, making the two directly comparable at matched memory. Momentum-free Adafactor is listed for reference.

Model	N	Adam	Adafactor-m	Adafactor	iKFAD	R-iKFAD
TinyViT	5.0M	30.7	15.6	0.2	30.7	15.6 ($2.0\times$)
DistilBERT	80.0M	510.8	256.1	0.7	510.8	256.1 ($2.0\times$)
GPT2-Nano	0.8M	6.1	3.1	< 0.1	6.1	3.1 ($2.0\times$)
GPT2-S	124.4M	948.9	475.4	0.9	948.9	475.4 ($2.0\times$)

D Accuracy Curves

Figure 7 gives the accuracy curves corresponding to the loss curves of Figure 3, for the two classification benchmarks. They are reported here rather than in Section 4 because they track the loss curves closely and add no separate conclusion.

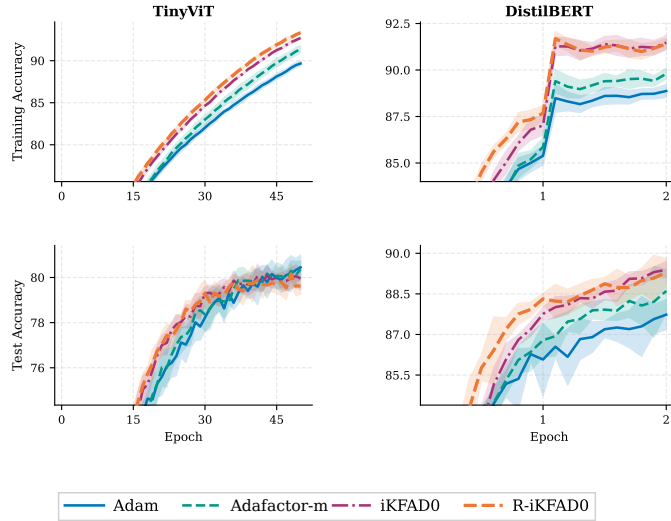


Figure 7: Accuracy averaged over 10 seeds. R-iKFAD0 matches iKFAD0 and Adam. Error bars denote one std.

E Prefactor Dependence

The prefactors depend on α , μ and ϵ_{stab} as the calculation of Section 3.1 predicts. If the asymptotics are $\mathcal{E} \sim c_1 \alpha \mu / t$ and $\mathcal{E} \sim c_2 \alpha \mu \sqrt{\epsilon_{\text{stab}} / 2t}$, then the ratios in Table 6 are approximately constant; they were evaluated at $t = 10^4$ with $m = 3$, $n = 2$. The crossover row locates the last downward crossing of $\mathbf{1}^\top R = \epsilon_{\text{stab}}$.

Table 6: Parameter dependence of the prefactors, evaluated at $t = 10^4$ with $m = 3, n = 2$. In the first row α and μ each range over a factor of 4, so $\alpha\mu$ ranges over a factor of 16, yet the ratio moves by less than 30%.

quantity	parameters varied	mean	spread (max/min)
$t\mathcal{E}/(\alpha\mu), \epsilon_{\text{stab}} = 0$	$\alpha, \mu \in \{\frac{1}{2}, 1, 2\}$	4.69	1.29
$\mathcal{E}\sqrt{t}/(\alpha\mu\sqrt{\epsilon_{\text{stab}}/2}), \epsilon_{\text{stab}} > 0$	$\alpha, \mu \in \{\frac{1}{2}, 1\}, \epsilon_{\text{stab}} \in \{0.05, 0.2\}$	2.31	1.14
$\mathcal{E}/(\alpha\mu\epsilon_{\text{stab}})$ at the crossing	$\alpha, \mu \in \{\frac{1}{2}, 1\}, \epsilon_{\text{stab}} \in \{0.05, 0.2\}$	0.87	1.21

The third row places the transition between the two regimes at $\mathcal{E} \approx 0.87 \alpha\mu \epsilon_{\text{stab}}$, confirming the crossover scale. For the scalar system $m = n = 1$ with $\alpha = \mu = 1$, evaluating at $t = 10^5$ gives $t\mathcal{E}/(\alpha\mu) = 0.909$ and $\mathcal{E}\sqrt{t}/(\alpha\mu\sqrt{\epsilon_{\text{stab}}/2}) = 0.884$, both within about 10% of the value 1 the averaging calculation gives; for $m, n > 1$ the same relations hold with a dimension-dependent constant, which that calculation does not supply.

F Numerical Setup for Section 3.2

The initial entries of X and P were independent normal variables with standard deviations 0.4 and 0.3, and $R(0) = C(0) = 0$ throughout. The parameter sweeps used MATLAB R2026a, `rng(0)`, and `ode89` with relative tolerance 10^{-10} and absolute tolerance 10^{-12} ; no maximum step was specified. When $\epsilon_{\text{stab}} = 0$ and $\mathbf{1}^\top R = 0$ the code set $\tilde{\xi} = 0$, and used (15) once $\mathbf{1}^\top R > 0$. The curves in Figure 2 were computed by BACD splitting with $m = 4, n = 3, \alpha = \mu = 1, h = 0.02$, final time 10^7 and random seed 1; the additional runs behind Table 1 used dimensions 2×2 and 6×5 with the same $A, X(0)$ and $P(0)$. No fitting uncertainties were recorded. As consistency checks, for $m = 3, n = 2$ and $\alpha = \mu = 1$, the largest absolute difference between the `ode89` and BACD slopes was 0.018 over the three fitted intervals. Repeating the six BACD fits at $h = 0.01$ instead of $h = 0.02$ changed each slope by less than 8.9×10^{-4} .

G Rank-1 iKFAD Convergence Analysis

We analyse the continuous-time Rank-1 iKFAD dynamics for a matrix parameter $X \in \mathbb{R}^{m \times n}$ (the vector case follows by flattening). The system is given by

$$\dot{X} = P, \quad (19a)$$

$$\dot{P} = -\nabla f(X) - \tilde{\xi} \odot P - \gamma P, \quad (19b)$$

$$\dot{R} = \frac{1}{\mu} [P]^2 \mathbf{1}_n - \alpha R, \quad (19c)$$

$$\dot{C} = \frac{1}{\mu} ([P]^2)^\top \mathbf{1}_m - \alpha C, \quad (19d)$$

where $P \in \mathbb{R}^{m \times n}, R \in \mathbb{R}^m, C \in \mathbb{R}^n$, and $\gamma \geq 0, \alpha, \mu > 0$ are hyperparameters. Boundedness (Lemma 4) holds for $\gamma \geq 0$; the exponential convergence results below additionally require $\gamma > 0$. The rank-1 friction tensor $\tilde{\xi} \in \mathbb{R}^{m \times n}$ is defined element-wise as

$$\tilde{\xi}_{ij} = \frac{R_i C_j}{\sum_{k=1}^m R_k + \epsilon_{\text{stab}}},$$

with $\epsilon_{\text{stab}} > 0$ a small constant preventing division by zero. The notation $\odot$ denotes element-wise multiplication, $[\cdot]^2$ element-wise squaring, and $\langle \cdot, \cdot \rangle_F$ the Frobenius inner product.

G.1 Dynamical properties and boundedness

Throughout this appendix X^* denotes a global minimiser of f . Under the hypotheses used below such a point exists, because f is continuous and coercive, and it satisfies

$$\nabla f(X^*) = 0 \quad \text{and} \quad f(X) \geq f(X^*) \quad \text{for all } X. \quad (20)$$

Both properties are used below, so we state them explicitly.

Lemma 4 (Boundedness of solutions). *Assume $f \in C^2$ (so ∇f is automatically locally Lipschitz), $f(X) \rightarrow +\infty$ as $\|X\|_F \rightarrow +\infty$, and let X^* be a global minimiser of f as in (20). Then for any initial condition (X_0, P_0, R_0, C_0)*

with $R_0, C_0 \geq 0$ entrywise, the solution of (19a)–(19d) is well-defined for all $t \geq 0$, and there exists a compact set $\mathcal{K}$ containing the trajectory. In particular, with

$$P_{\max}^2 := 2(f(X_0) - f(X^*) + \tfrac{1}{2}\|P_0\|_F^2), \quad R_{\max} := \|R_0\| + \frac{P_{\max}^2}{\alpha\mu}, \quad C_{\max} := \|C_0\| + \frac{P_{\max}^2}{\alpha\mu},$$

all of which are finite and independent of t , we have for all $t \geq 0$

$$0 \leq R_i(t) \leq R_{\max}, \quad 0 \leq C_j(t) \leq C_{\max}, \quad \|P(t)\|_F \leq P_{\max},$$

and moreover the friction tensor obeys the bound

$$0 \leq \tilde{\xi}_{ij}(t) \leq \xi_{\max} := \min\left\{C_{\max}, \frac{R_{\max}C_{\max}}{\epsilon_{\text{stab}}}\right\}. \quad (21)$$

Note that $P_{\max}$ depends on neither μ nor γ .

Proof. The ODEs (19c)–(19d) have non-negative forcing and linear decay, so $R_0, C_0 \geq 0$ implies $R_i(t), C_j(t) \geq 0$ for all $t \geq 0$, and hence $\tilde{\xi}_{ij} \geq 0$ throughout.

Consider the reduced energy

$$\mathcal{E}(X, P) = f(X) - f(X^*) + \tfrac{1}{2}\|P\|_F^2,$$

which omits the R, C contributions. Differentiating along trajectories and cancelling the gradient terms,

$$\dot{\mathcal{E}} = \langle \nabla f(X), P \rangle_F + \langle P, -\nabla f(X) - \tilde{\xi} \odot P - \gamma P \rangle_F = -\gamma\|P\|_F^2 - \sum_{i,j} \tilde{\xi}_{ij} P_{ij}^2 \leq 0, \quad (22)$$

both terms being non-positive because $\gamma \geq 0$ and $\tilde{\xi} \geq 0$ entrywise. No bound on $\|P\|_F$ is used in deriving (22), so the estimate is not circular. Consequently $\mathcal{E}(t) \leq \mathcal{E}(0)$ for all t , which yields at once

$$\|P(t)\|_F^2 \leq 2\mathcal{E}(0) =: P_{\max}^2,$$

and, since $f(X) - f(X^*) \leq \mathcal{E}(0)$ with f coercive, confinement of $X(t)$ to a compact sublevel set of f .

With $\|P\|_F \leq P_{\max}$ now established independently, the factors follow from the explicit linear filter. Integrating (19c),

$$R(t) = e^{-\alpha t} R_0 + \frac{1}{\mu} \int_0^t e^{-\alpha(t-s)} [P(s)]^2 \mathbf{1}_n \, ds,$$

and using the sharp, dimension-free estimate $\|[P]^2 \mathbf{1}_n\| \leq \|P\|_F^2 \leq P_{\max}^2$ (the entries of $[P]^2 \mathbf{1}_n$ are non-negative and sum to $\|P\|_F^2$, so its 2-norm is at most its 1-norm),

$$\|R(t)\| \leq \|R_0\| + \frac{P_{\max}^2}{\alpha\mu} =: R_{\max},$$

with the analogous bound $C_{\max} := \|C_0\| + P_{\max}^2/(\alpha\mu)$. All four variables therefore remain in a compact set $\mathcal{K}$ for all $t \geq 0$, and global existence follows from the standard continuation criterion for locally Lipschitz vector fields.

Finally we establish (21). Two bounds are available. Bounding the denominator below by ϵ_{stab} gives $\tilde{\xi}_{ij} \leq R_{\max}C_{\max}/\epsilon_{\text{stab}}$, which diverges as $\epsilon_{\text{stab}} \downarrow 0$. The second is sharper and uniform in ϵ_{stab} : since every $R_k \geq 0$ we have $R_i \leq \mathbf{1}_m^\top R$, hence

$$\tilde{\xi}_{ij} = \frac{R_i}{\mathbf{1}_m^\top R + \epsilon_{\text{stab}}} C_j \leq C_j \leq C_{\max},$$

and (21) is the minimum of the two. The constant $\epsilon_{\text{stab}} > 0$ remains necessary for well-posedness at $R = 0$, which is exactly how the implementation initialises the factors, but it does not enter the bound on $\tilde{\xi}$, and hence does not enter the convergence rate. $\square$

G.2 Geometric convergence

Theorem 1. Suppose that f is m_f -strongly convex and has an L_f -Lipschitz gradient. Let $\gamma, \alpha, \mu, \epsilon_{\text{stab}} > 0$, and let $R(0)$ and $C(0)$ be componentwise nonnegative. Then there are constants $M, \rho > 0$ such that

$$f(X(t)) - f(X^*) + \|X(t) - X^*\|_F^2 + \|P(t)\|_F^2 + \|R(t)\|^2 + \|C(t)\|^2 \leq M e^{-\rho t} \quad (23)$$

for every $t \geq 0$.

Proof of Theorem 1. Put $Y = X - X^*$ and $d_{\max} = \gamma + \xi_{\max}$. Choose

$$0 < \delta \leq \min \left\{ \frac{\sqrt{m_f}}{2}, \frac{\gamma}{2(1 + d_{\max}^2/(2m_f))} \right\}. \quad (24)$$

Here $\xi_{\max}$ is the bound in Lemma 4. Define

$$V = f(X) - f(X^*) + \frac{1}{2}\|P\|_F^2 + \delta \langle Y, P \rangle_F.$$

Young's inequality gives

$$V \geq \frac{m_f}{4}\|Y\|_F^2 + \frac{1}{4}\|P\|_F^2. \quad (25)$$

It also gives

$$V \leq \left(\frac{L_f}{2} + \frac{m_f}{4} \right) \|Y\|_F^2 + \frac{3}{4}\|P\|_F^2. \quad (26)$$

Differentiating along the equations gives

$$\begin{aligned} \dot{V} &\leq -(\gamma - \delta)\|P\|_F^2 - \delta m_f \|Y\|_F^2 + \delta d_{\max} \|Y\|_F \|P\|_F \\ &\leq -\frac{\delta m_f}{2} \|Y\|_F^2 - \frac{\gamma}{2} \|P\|_F^2. \end{aligned} \quad (27)$$

The last line follows from Young's inequality and (24).

Set

$$\rho_0 := \min \left\{ \frac{2\delta m_f}{2L_f + m_f}, \frac{2\gamma}{3} \right\}.$$

Equations (26) and (27) imply

$$\dot{V} \leq -\rho_0 V.$$

Therefore $V(t) \leq e^{-\rho_0 t} V(0)$.

The lower bound (25) now gives exponential decay of $X - X^*$ and P . The variation-of-constants formulas for R and C then give their exponential decay. For example,

$$R_i(t) = e^{-\alpha t} R_i(0) + \frac{1}{\mu} \int_0^t e^{-\alpha(t-s)} \sum_{k=1}^n P_{ik}(s)^2 \, ds.$$

The convolution on the right decays exponentially. The same argument applies to C . Taking, for example,

$$0 < \rho < \min\{\rho_0, 2\alpha\}$$

and increasing M if necessary proves (17). The same bounds prevent finite-time escape, so the solution is defined for every $t \geq 0$. $\square$

The proof uses the fixed term $-\gamma\|P\|_F^2$ to control the position–momentum cross term. When $\gamma = 0$, this control is absent. The next subsection therefore treats this case separately.

G.3 The case $\gamma = 0$

In our experiments we observed that $\gamma = 0$ gave optimal or near-optimal results. The next theorem proves convergence of the continuous system in this case.

Theorem 2. Suppose that f is strongly convex and has a locally Lipschitz gradient. Let $\alpha, \mu, \epsilon_{\text{stab}} > 0$, and suppose that $R(0)$ and $C(0)$ are componentwise nonnegative. Then every solution of (19a)–(19d) with $\gamma = 0$ converges to

$$(X, P, R, C) = (X_*, 0, 0, 0),$$

where X_* is the unique minimiser of f .

Proof of Theorem 2. Set

$$\mathcal{H}(X, P) = f(X) + \frac{1}{2} \|P\|_F^2.$$

Along a solution with $\gamma = 0$,

$$\begin{aligned} \dot{\mathcal{H}} &= \langle \nabla f(X), P \rangle_F + \langle P, -\nabla f(X) - \tilde{\xi} \odot P \rangle_F \\ &= - \sum_{i,j} \tilde{\xi}_{ij} P_{ij}^2 \leq 0. \end{aligned} \quad (28)$$

Thus $\mathcal{H}$ is nonincreasing. Its sublevel sets are bounded by strong convexity. Lemma 4 gives global existence.

The factors satisfy

$$R_i(t) = e^{-\alpha t} R_i(0) + \frac{1}{\mu} \int_0^t e^{-\alpha(t-s)} \sum_{k=1}^n P_{ik}(s)^2 \, ds, \quad (29)$$

$$C_j(t) = e^{-\alpha t} C_j(0) + \frac{1}{\mu} \int_0^t e^{-\alpha(t-s)} \sum_{\ell=1}^m P_{\ell j}(s)^2 \, ds. \quad (30)$$

These formulas show that the factors remain nonnegative. Since P is bounded, they also show that R and C are bounded. The complete trajectory is therefore precompact.

It remains to identify the largest invariant subset of $\{\dot{\mathcal{H}} = 0\}$. Equation (28) and $\tilde{\xi}_{ij} \geq 0$ imply that

$$\tilde{\xi}_{ij} P_{ij}^2 = 0 \quad \text{for every } i, j$$

on this set. Suppose that $P_{ij} \neq 0$ at some point of a trajectory in the set. Continuity makes $P_{ij}^2 > 0$ on a time interval. Because $\epsilon_{\text{stab}} > 0$, the displayed equality then implies $R_i C_j = 0$ throughout that interval.

This is impossible. If $R_i = 0$ at any time in the interval, then

$$\dot{R}_i = \frac{1}{\mu} \sum_k P_{ik}^2 > 0.$$

Similarly, if $C_j = 0$, then

$$\dot{C}_j = \frac{1}{\mu} \sum_{\ell} P_{\ell j}^2 > 0.$$

Moreover, the variation-of-constants formulas show that a nonnegative factor, once positive, stays positive. Hence both R_i and C_j become positive in the interval, contradicting $R_i C_j = 0$. We conclude that every trajectory in $\{\dot{\mathcal{H}} = 0\}$ has $P = 0$.

If $P = 0$ along an invariant trajectory, then $\nabla f(X) = 0$ and $X = X^*$. The largest invariant subset of $\{\dot{\mathcal{H}} = 0\}$ is therefore

$$\{(X^*, 0, R, C) : R, C \geq 0\}.$$

LaSalle's invariance principle gives

$$X(t) \longrightarrow X^*, \quad P(t) \longrightarrow 0.$$

Finally, equations (29)–(30) and $P(t) \rightarrow 0$ give $R(t), C(t) \rightarrow 0$. □

G.3.1 Algebraic rates

The formal calculation above identifies the exponents. The next result establishes them rigorously, with matching upper and lower bounds in both regimes.

Throughout this subsection $\gamma = 0$ in (19a)–(19d), and the friction is

$$\tilde{\xi}_{ij} = \frac{R_i C_j}{s + \epsilon_{\text{stab}}}, \quad s = \sum_i R_i. \quad (31)$$

When $\epsilon_{\text{stab}} = 0$, set $\tilde{\xi} = 0$ at $s = 0$. Since $\dot{R}_i \geq -\alpha R_i$ and $\dot{C}_j \geq -\alpha C_j$, nonnegative initial factors remain nonnegative, so $s \geq 0$ along the flow. We write $\|\tilde{\xi}\|_{\infty} = \max_{i,j} |\tilde{\xi}_{ij}|$.

Let X^* be the minimiser of f , and define

$$\mathcal{E}(t) = f(X(t)) - f(X^*) + \frac{1}{2} \|P(t)\|_F^2. \quad (32)$$

Theorem 3. Suppose that f is strongly convex, its gradient is locally Lipschitz, and it is C^2 in a neighbourhood of X^* . Let $\alpha, \mu > 0$, and let the initial factors be componentwise nonnegative.

If $\epsilon_{\text{stab}} > 0$ and $\mathcal{E}(0) > 0$, there are $a, A > 0$ such that

$$\frac{a}{\sqrt{1+t}} \leq \mathcal{E}(t) \leq \frac{A}{\sqrt{1+t}}, \quad t \geq 0.$$

If $\epsilon_{\text{stab}} = 0$, assume in addition that $R(0) = C(0) = 0$, and use the convention following (31). If $\mathcal{E}(0) > 0$, there are $b, B > 0$ such that

$$\frac{b}{1+t} \leq \mathcal{E}(t) \leq \frac{B}{1+t}, \quad t \geq 0.$$

The proof uses two estimates. A lower bound on the energy lost over a fixed time window gives the upper bound on $\mathcal{E}$. An upper bound on the instantaneous loss gives the lower bound on $\mathcal{E}$.

Lemma 5 (The zero-stabiliser phase space). *Suppose that $\epsilon_{\text{stab}} = 0$ and $\sum_i R_i = \sum_j C_j =: s$. The convention $\tilde{\xi} = 0$ at $s = 0$ defines a locally Lipschitz friction tensor on*

$$\mathcal{B} = \left\{ R \geq 0, C \geq 0, \sum_i R_i = \sum_j C_j \right\}.$$

Moreover,

$$0 \leq \tilde{\xi}_{ij} \leq C_j \leq s. \quad (33)$$

Proof. The bound follows from $R_i \leq s$. For local Lipschitz continuity, consider two points of $\mathcal{B}$ with totals $0 \leq s \leq s'$. If $s = s' = 0$, both friction tensors vanish. If $s = 0 < s'$, then

$$|\tilde{\xi}_{ij} - \tilde{\xi}'_{ij}| = \frac{R'_i C'_j}{s'} \leq s' = |s - s'|.$$

If $s > 0$, write

$$R = s\rho, \quad C = s\sigma, \quad R' = s'\rho', \quad C' = s'\sigma',$$

where the four normalised vectors lie in the appropriate simplices. Then

$$|s\rho_i\sigma_j - s'\rho'_i\sigma'_j| \leq |s - s'| + s|\rho_i - \rho'_i| + s|\sigma_j - \sigma'_j|,$$

and

$$s|\rho_i - \rho'_i| \leq |R_i - R'_i| + |s - s'|, \quad s|\sigma_j - \sigma'_j| \leq |C_j - C'_j| + |s - s'|.$$

Since $|s - s'| \leq \|R - R'\|_1$, these estimates give the required Lipschitz bound. $\square$

Lemma 6 (Kinetic energy on a fixed window). *Assume the hypotheses of Theorem 3, and suppose a solution of (19a)–(19d) satisfies*

$$X(t) \longrightarrow X^*, \quad P(t) \longrightarrow 0, \quad \|\tilde{\xi}(t)\|_\infty \longrightarrow 0.$$

Then for every $T > 0$ there are $c_0 > 0$ and $t_0 \geq 0$, both depending on the solution, such that

$$\int_t^{t+T} \|P(u)\|_F^2 du \geq c_0 \mathcal{E}(t) \quad (t \geq t_0). \quad (34)$$

Proof. Fix $T > 0$. If the result is false, for every k there is $t_k \geq k$ violating (34) with $c_0 = 1/k$. The integral is nonnegative, so $e_k = \mathcal{E}(t_k) > 0$, and

$$\frac{1}{e_k} \int_{t_k}^{t_k+T} \|P(u)\|_F^2 du \longrightarrow 0. \quad (35)$$

Set, for $0 \leq u \leq T$,

$$y_k(u) = \frac{X(t_k + u) - X^*}{\sqrt{e_k}}, \quad p_k(u) = \frac{P(t_k + u)}{\sqrt{e_k}}.$$

The assumed convergence gives $e_k \rightarrow 0$. Energy monotonicity and strong convexity make y_k and p_k uniformly bounded. On bounded sets of y , the C^2 assumption gives, uniformly,

$$\frac{\nabla f(X^* + \sqrt{e_k}y)}{\sqrt{e_k}} \rightarrow Hy, \quad H = \nabla^2 f(X^*).$$

The equations and the uniform convergence of $\tilde{\xi}(t_k + u)$ to zero on $[0, T]$ now give uniform bounds on $\dot{y}_k$ and $\dot{p}_k$. Hence there is a constant L , independent of k , such that

$$\|y_k(u) - y_k(v)\|_F + \|p_k(u) - p_k(v)\|_F \leq L|u - v| \quad (0 \leq u, v \leq T).$$

The Arzelà–Ascoli theorem therefore gives a subsequence on which both sequences converge uniformly, say to y and p . For this subsequence, the equations are

$$\begin{aligned} y_k(u) &= y_k(0) + \int_0^u p_k(v) dv, \\ p_k(u) &= p_k(0) - \int_0^u \left[\frac{\nabla f(X^* + \sqrt{e_k}y_k(v))}{\sqrt{e_k}} + \tilde{\xi}(t_k + v) \odot p_k(v) \right] dv. \end{aligned}$$

Uniform convergence gives $p_k \rightarrow p$. The Taylor estimate above gives uniform convergence of the scaled gradient to Hy , while $\tilde{\xi}(t_k + v) \odot p_k(v) \rightarrow 0$ uniformly. Taking the limit in these identities gives

$$y(u) = y(0) + \int_0^u p(v) dv, \quad p(u) = p(0) - \int_0^u Hy(v) dv.$$

Thus the limits solve

$$\dot{y} = p, \quad \dot{p} = -Hy. \quad (36)$$

Taylor expansion of f at X^* gives

$$\frac{1}{2} \langle Hy(0), y(0) \rangle + \frac{1}{2} \|p(0)\|_F^2 = 1.$$

Equation (35) gives $\int_0^T \|p(u)\|_F^2 du = 0$. Thus $p = 0$ on $[0, T]$. Equation (36) gives $Hy = 0$. Strong convexity makes H positive definite, so $y = 0$, contradicting the normalised energy. $\square$

Proof of Theorem 3. Boundedness and convergence. Differentiating (32) gives

$$\dot{\mathcal{E}}(t) = - \sum_{i,j} \tilde{\xi}_{ij}(t) P_{ij}(t)^2. \quad (37)$$

Thus $\mathcal{E}$ is nonincreasing. Strong convexity bounds X and P . Variation of constants gives

$$R_i(t) = e^{-\alpha t} R_i(0) + \frac{1}{\mu} \int_0^t e^{-\alpha(t-u)} \sum_j P_{ij}(u)^2 du, \quad (38)$$

$$C_j(t) = e^{-\alpha t} C_j(0) + \frac{1}{\mu} \int_0^t e^{-\alpha(t-u)} \sum_i P_{ij}(u)^2 du. \quad (39)$$

Hence the factors are also bounded. Local Lipschitz continuity, together with boundedness, gives global existence. In the zero-stabiliser case, local Lipschitz continuity is understood on the phase space in Lemma 5. If $\mathcal{E}(0) = 0$, then $X = X^*$, $P = 0$, and the energy remains zero. Assume from now on that $\mathcal{E}(0) > 0$.

Consider a complete bounded trajectory in the ω -limit set on which $\dot{\mathcal{E}} = 0$. Boundedness of the complete trajectory gives

$$R_i(t) = \frac{1}{\mu} \int_{-\infty}^t e^{-\alpha(t-u)} \sum_j P_{ij}(u)^2 du, \quad C_j(t) = \frac{1}{\mu} \int_{-\infty}^t e^{-\alpha(t-u)} \sum_i P_{ij}(u)^2 du.$$

If some $P_{ij} \neq 0$, it remains nonzero on a short interval, and these formulas make both R_i and C_j positive on a nonempty subinterval. This makes $\tilde{\xi}_{ij} P_{ij}^2 > 0$, a contradiction. Hence $P = 0$ on every such trajectory. Invariance then gives $\nabla f(X) = 0$, and thus $X = X^*$. LaSalle's invariance principle and the filter formulas give

$$X(t) \rightarrow X^*, \quad P(t) \rightarrow 0, \quad R(t), C(t) \rightarrow 0. \quad (40)$$

For $\epsilon_{\text{stab}} = 0$, the two factor totals satisfy the same scalar equation. The initialization therefore keeps the solution in $\mathcal{B}$, and Lemma 5 applies. In both cases, (40) gives $\|\tilde{\xi}(t)\|_\infty \rightarrow 0$. Since the friction is also bounded, $\|\tilde{\xi}\|_\infty \leq M_0$ for a fixed M_0 , so (37) gives $-\dot{\mathcal{E}} \leq M_0 K \leq 2M_0 \mathcal{E}$, where $K = \|P\|_F^2$. Hence $\mathcal{E}(t) \geq \mathcal{E}(0)e^{-2M_0 t} > 0$.

The factor totals. Let q denote either $s = \sum_i R_i$ or $\bar{s} = \sum_j C_j$. Both satisfy

$$\dot{q} = \frac{K}{\mu} - \alpha q.$$

Set $w = q/\mathcal{E}$. Since $K \leq 2\mathcal{E}$, equation (37) gives

$$\dot{w} \leq \frac{2}{\mu} - \alpha w + 2\|\tilde{\xi}\|_\infty w.$$

The last coefficient tends to zero. Scalar comparison therefore gives constants $Z_R, Z_C > 0$ such that, for all sufficiently large t ,

$$s(t) \leq Z_R \mathcal{E}(t), \quad \bar{s}(t) \leq Z_C \mathcal{E}(t). \quad (41)$$

Lemma 6 also applies. Choose t_1 so that (34) and (41) hold for $t \geq t_1$.

A common window estimate. Let $d = mn$. For each $t \geq t_1$, choose (i, j) such that

$$G := \int_t^{t+T} P_{ij}(u)^2 du \geq \frac{c_0}{d} \mathcal{E}(t). \quad (42)$$

For $t \leq u \leq t+T$, put

$$G(u) = \int_t^u P_{ij}(v)^2 dv,$$

so that $G(t) = 0$ and $G(t+T) = G$. Equations (38) and (39), with their nonnegative omitted terms, give

$$R_i(u) \geq \frac{e^{-\alpha T}}{\mu} G(u), \quad C_j(u) \geq \frac{e^{-\alpha T}}{\mu} G(u).$$

It follows that

$$\begin{aligned} \int_t^{t+T} R_i(u) C_j(u) P_{ij}(u)^2 du &\geq \frac{e^{-2\alpha T}}{\mu^2} \int_t^{t+T} G(u)^2 G'(u) du \\ &= \frac{e^{-2\alpha T}}{3\mu^2} G^3 \geq c_1 \mathcal{E}(t)^3, \end{aligned} \quad (43)$$

where $c_1 > 0$ is fixed.

Upper bound when $\epsilon_{\text{stab}} > 0$. Boundedness of R gives $s(u) + \epsilon_{\text{stab}} \leq M$ for a fixed M . Integrating (37), retaining the single index chosen above, and using (43) yields

$$\mathcal{E}(t) - \mathcal{E}(t+T) \geq c_2 \mathcal{E}(t)^3. \quad (44)$$

Upper bound when $\epsilon_{\text{stab}} = 0$. The solution is not the equilibrium, and (38) gives $s(t) > 0$ for every $t > 0$. Throughout a window $[t, t+T]$ with $t \geq t_1$, (41) and monotonicity give

$$s(u) \leq Z_R \mathcal{E}(u) \leq Z_R \mathcal{E}(t).$$

Equations (37) and (43) now give

$$\mathcal{E}(t) - \mathcal{E}(t+T) \geq c_3 \mathcal{E}(t)^2. \quad (45)$$

Iteration of the window estimates. Set $e_k = \mathcal{E}(t_1 + kT)$. If some $e_k = 0$, then $\mathcal{E} \equiv 0$ and the bound is trivial, so assume $e_k > 0$ for every k . If

$$e_k - e_{k+1} \geq c e_k^q,$$

then the mean value theorem gives

$$e_{k+1}^{-(q-1)} - e_k^{-(q-1)} \geq (q-1)c.$$

Thus $e_k = O(k^{-1/(q-1)})$. Monotonicity of $\mathcal{E}$ extends the bound from the grid points to every $t \geq t_1$. Increasing the constant covers $[0, t_1]$. Taking $q = 3$ in (44) and $q = 2$ in (45) gives the two upper bounds.

Lower bound when $\epsilon_{\text{stab}} = 0$. By (33), (37), and (41),

$$-\dot{\mathcal{E}} \leq sK \leq 2Z_R \mathcal{E}^2, \quad \text{that is,} \quad \frac{d}{dt} \frac{1}{\mathcal{E}} \leq 2Z_R,$$

for $t \geq t_1$. Integrating from t_1 gives

$$\frac{1}{\mathcal{E}(t)} \leq \frac{1}{\mathcal{E}(t_1)} + 2Z_R(t - t_1), \quad t \geq t_1,$$

and hence a constant $b_1 > 0$ with $\mathcal{E}(t) \geq b_1/(1+t)$ for $t \geq t_1$.

Lower bound when $\epsilon_{\text{stab}} > 0$. Since $R_i \leq s$, $C_j \leq \bar{s}$, and $K \leq 2\mathcal{E}$, (37) and (41) give

$$\tilde{\xi}_{ij} \leq \frac{s\bar{s}}{\epsilon_{\text{stab}}}, \quad -\dot{\mathcal{E}} \leq \frac{s\bar{s}}{\epsilon_{\text{stab}}} K \leq \frac{2Z_R Z_C}{\epsilon_{\text{stab}}} \mathcal{E}^3, \quad \text{that is,} \quad \frac{d}{dt} \frac{1}{\mathcal{E}^2} \leq \frac{4Z_R Z_C}{\epsilon_{\text{stab}}},$$

for $t \geq t_1$. Integrating from t_1 gives

$$\frac{1}{\mathcal{E}(t)^2} \leq \frac{1}{\mathcal{E}(t_1)^2} + \frac{4Z_R Z_C}{\epsilon_{\text{stab}}}(t - t_1), \quad t \geq t_1,$$

and hence a constant $a_1 > 0$ with $\mathcal{E}(t) \geq a_1/\sqrt{1+t}$ for $t \geq t_1$. Reducing a_1 and b_1 , if necessary, extends both lower bounds to $[0, t_1]$. $\square$